CELESTIAL MECHANICS × AI · 2026

From the next token to the next theorem

How a model works — and how I use it in practice



Bowen Liu

What the model sees: ten integers

EN · ONE SENTENCE

Celestial mechanics is the poetry of orbits.

Cel	estial	_mechanics	_is	_the	_poetry	_of	_or	bits	.
69126	90010	52347	382	290	38153	328	503	24043	13

46 characters → **10** tokens → 10 integers · _ marks a leading space

CHINESE · FOUNDATIONS OF CELESTIAL MECHANICS

本书系统地介绍了天体力学

本	书	系统	地	介绍	了	天	体	力	学
4087	20121	31936	5557	37887	4531	867	7074	11343	8590

12 characters → **10** tokens → 10 integers · two-character words often map to one token

“This book systematically introduces celestial mechanics.”

01 Where do these integers come from?

02 How does it predict the next one?

03 How do we put it to work?

Put the model back in the system



Inner ring: one call · Middle ring: across calls · Outer ring: engineering boundary

01	**LLM**	A probability generator: tokens in, a distribution over the next token out
02	**Prompt**	This call's conditions: system / user / few-shot
03	**Context**	Every token visible to this call; the window is finite
04	**Memory**	State beyond the window: write / retrieve / compress / forget
05	**Agent loop**	Observe → act → observe; state lives outside
06	**Harness**	Tools, sandbox, permissions, verification, logs

> The model is the CPU; prompt, context, memory, loop and harness are the runtime.

Acceleration: from basic use to gold medals

2023 · BEFORE

Basic use

- Explain concepts, search the literature, write code
- Do simple calculations, often wrong
- Complex proofs still need people
- Role: assistant

→ 19 months

2025 · AFTER

Research level

- IMO 2025 gold level 35/42
- Lean proofs, machine-checkable
- OpenAI 2026: 372 problems, ≈3 h each (report)
- Role: candidate generator

2024.07 **28/42** silver → 2025.07 **35/42** gold level 12 months, +7 points

> This is a staircase with **three steps**.

Timeline: three steps

2023–24

Tools: search + verification

2023.12
FunSearch
LLM + evaluator + evolutionary search; a new cap-set construction

2024.01
AlphaGeometry
Neuro-symbolic; IMO geometry 25/30

2024.07
AlphaProof
Lean + RL; IMO 2024 silver 28/42 (+AG2)

2024–25

Reasoning: RL + test-time compute

2024.09
OpenAI o1
RL teaches thinking; test-time compute

2025.01
DeepSeek-R1
RLVR; reasoning behavior emerges; open weights

2025.05
AlphaEvolve
Evolutionary coding agent; 4×4 complex matrix product in 48 multiplications

2025–26

Scale: from benchmarks to open problems

2025.07
IMO 2025
Gold level 35/42; natural language, no tools

2025
Formal gold (claimed)
Aristotle et al.; Lean-checkable

2026.05–10
OpenAI
Unit-distance counterexample; NS / Riemann-related (claimed); 372 problems (report)

> Every leap combines a **generator, a verifier and search**.

The motor and the assembly line

A story Eric Schmidt tells — and the AI parallel.

When the electric motor was invented, factory owners did the obvious thing: remove the central steam engine, put a motor in its place, and change nothing else. Efficiency rose by about 30%.

Decades later, someone realized the motor could be small. Every machine could have its own. The factory could be designed around the flow of production itself; the power source no longer dictated the layout.

The assembly line was born. Efficiency rose by an order of magnitude.

30%

Standalone · Prompting Harder
Resets each session · No compounding

10×+

Assembly line · Context Engineering
Persistent assets & rules · Continuous compounding

The gap is **work designed to compound**.

> Models are rented; context and systems are owned. Do not merely prompt in isolation; turn domain rules, tools, and evals into persistent assets.

> The gain is in **layout and compounding**.

From the chat box to the working directory

Research cannot stay in thin wrappers. Files are interfaces; rules are portable.

L1 · WRAPPER

Web chat

Ask, read, manual copy
No local file access · Ephemeral context

Zero real tools
The model suggests; the human executes

L2–3 · COPILOT

IDE editor

In-file completions
Context limited to currently open files

A faster typewriter
Same human loop, slightly higher cadence

L4–5 · AGENT LOOP

Repo & CLI Agent

Tools & isolated agents
Symplectic integrators · Series expansion · Reviewers

Autonomous verification
Hypothesis → Experiment → Conservation checks → git

- OpenCode
- Claude Code
- Codex CLI
- Cursor
- Markdown SOP Protocol

> Files are interfaces. Build **portable research assets** in local workspaces.

`Swap the foundation model via a single config line; your dynamical check suites remain permanently valid.`

Two attempts

Same intention. Different governance: from abdication to coaching.

OPENCLAW · EARLY ATTEMPT

Mistaken for calculator · Hands-off

- Expected the agent to autonomously deliver an end-to-end paper.
- Early model was brittle; hallucinations compounded without evals.
- Complete human abdication, hoping for an unguided miracle.

Lesson: unguided execution collapses deterministically.

CODEX · GPT-5.3–5.5 · ISTART

Act as Coach · Verification loop

- Postdoc extension; I framed clear goals (WHAT) while AI drafted (HOW).
- Refused to be a line-by-line fixer; coached by flagging flawed steps.
- Caught absurd derivation: $Ax = b,\ b \neq 0 \rightarrow x = 0$ ✕
- After 3 rejections, GPT-6.1 Sol scanned deeper issues for final fixes.

Lesson: verification asymmetry — evaluating is easier than drafting.

> Do not act as a line-by-line fixer; act as a **coach defining standards and counterexamples**.

Low-hanging fruit & compounding assets

Drafting a paper is only the start; compounding assets are evals and taste.

01 · HUMAN AS COACH

Frame WHAT & bounds

Define clear objectives; let AI explore the HOW. Point out logical flaws rather than patching code by hand.

02 · SYSTEMATIC EVALS

Dynamical checks

Models get stuck. Build invariant test suites: Hamiltonian conservation, symplectic symmetry, and residuals.

03 · COMPOUNDING ASSETS

Taste & eval suites

A single paper depreciates quickly. What compounds is your knowledge base, codebase SOPs, and formal scorers.

OpenAI can brute-force batches of open problems, but old conjectures were written for humans. As marginal output cost hits zero, only **eval systems and contrarian taste** compound.

> The central question: which problems are **worth tackling together with machines**?

From the next token to the next theorem

`token` → `probability` → `candidate` → `verification` → **`theorem`**

Probability gives breadth

Verification gives correctness

People set the direction

> The next theorem still begins with the **next token**.